

CHIMERA-QRC: A Pre-Registered, Adversarially-Controlled Study of Quantum Reservoir Computing for S&P 500 Realized-Volatility Forecasting

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1. Focus, track selection, and problem framing

We target **Track A: financial volatility** — one-step-ahead forecasting of S&P 500 **realized variance (RV) and the tracking of volatility** regime transitions, the domain where JonesTrading's risk, allocation and derivatives pricing live. RV is long-memory, multi-scale, regime-switching and non-Gaussian — the structure reservoir computing exploits. Our central yardsticks are the Echo State Network (ESN, the classical analog of QRC) and the strongest econometric models (HAR — *Heterogeneous AutoRegressive*, Corsi 2009 — and, critically for Phase 3, **HAR-X**: HAR augmented with the same realized-measure features we feed the quantum reservoir). We also report GARCH/GJR-GARCH, AR(3), persistence, LSTM, and an RFF/RBF kernel, and we implement the challenge's common **MNIST** benchmark on the identical engine.

This paper's honest thesis. We pre-registered a falsifiable advantage hypothesis (H0) and then *attacked our own result* with the strongest fair controls. The finding is a **carefully bounded negative: at simulable scale (≤ 16 qubits) the quantum reservoir shows no statistically significant forecasting advantage** over strong classical baselines once they are given the same information; the dominant lever is *which features are encoded*, not the reservoir's quantum nonlinearity. We report what *does* survive — measurable kernel distinctness, lower seed variance, and full-rank entanglement (classical-simulation hardness) — as quantum-specific *properties* that are necessary, not sufficient, for advantage.

Vs. prior work. The closest study (Li et al. 2025/26 — QRC for realized volatility) presents the approach as a competitive, noise-resilient *proof-of-concept*; we add the control-hardened, pre-registered evaluation it omits — the decisive **HAR-X** control, HAC-DM/Holm/MCS, and explicit capability and per-layer-noise audits — turning an encouraging proof-of-concept into a falsifiable result.

2. QRC architecture (Fig. 1)

CHIMERA is a delay-embedding quantum reservoir. Inputs are angle-encoded $R_Y(\pi \cdot x)$, one value per qubit, onto $|0\dots 0\rangle$; the state evolves under a fixed transverse-field **Ising** Hamiltonian $H = \sum_{\{i<j\}} J_{ij} Z_i Z_j + h_x \sum_i X_i$ ($h_x=1$; J a **random $\approx 50\%$ -connected** graph, $\text{connectivity}=0.5$; $U=\exp(-iH\tau)$, $\tau=2$); single- and pairwise Pauli-Z expectations $\langle Z_i \rangle$, $\langle Z_i Z_j \rangle$ form the feature vector. A **ridge** head consumes these features concatenated with a linear block of the same inputs (incl. HAR), so any reservoir gain is genuine nonlinearity *beyond* the linear span of identical information. The identical inputs feed the classical controls, isolating quantum-vs-classical. Four mechanisms extend the core (multi-scale τ -bank; RZ measurement feedback; regime-adaptive Ising $\leftrightarrow$ Heisenberg via BOCPD; dissipation-as-feature); Phase 3 adds an **encoding-density / data-re-uploading path** (§5.2) and a **sparse/tensor backend** (§5.5). The engine is pure NumPy; an explicit PennyLane circuit reproduces it to $\approx 5 \times 10^{-4}$ and compiles, at $n=8$, to **380 native gates** (220 two-qubit + 160 one-qubit, 20 Trotter layers) — consistent with the $\approx 50\%$ -connected graph.

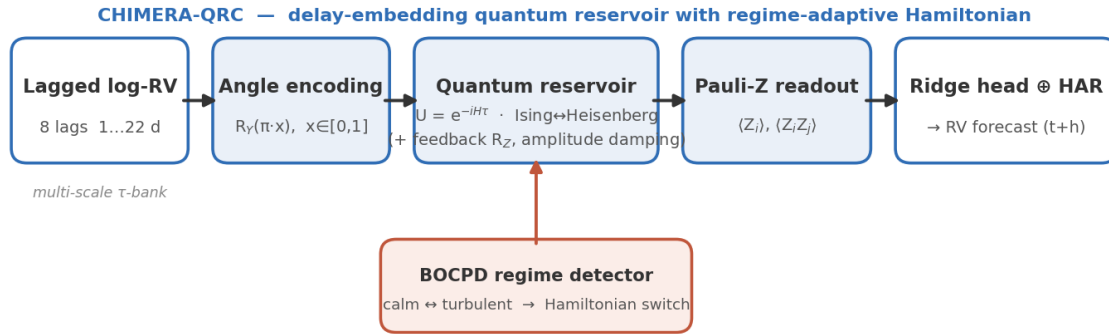


Figure 1. CHIMERA-QRC pipeline: lagged log-RV (and, in Axis B, realized-measure) inputs are angle-encoded onto qubits, evolved under a fixed random-coupled Ising Hamiltonian, and read out as single/pairwise Pauli-Z expectations into a ridge head fused with a linear block of the same inputs; a BOCPD detector selects the Hamiltonian for regime adaptivity.

3. Theoretical and analytical justification

The exploited quantum resources map to RV's structure: (a) **Hilbert-space dimensionality** —

n qubits give 2^n amplitudes and $O(n^2)$ measured Pauli features at $O(n^2)$ parameter cost;

(b) **intrinsic nonlinearity** from fixed unitary evolution (no gradient training, no barren plateaus); (c) **fading memory** (echo-state property); (d) **Hamiltonian as inductive bias**.

The central, falsifiable mechanism is **expressivity scaling**, which we measure directly via the geometric difference (Huang et al. 2021). The quantum kernel is poorly reproducible by a matched classical reservoir: against the name-matched ESN-108 reference,

$g(\text{ESN} \rightarrow \text{CHIMERA}) \approx 62$ vs ≈ 4 control (`cli.py run kernel`); against per- n *feature-matched* ESNs the gap is larger still ($g \approx 125$ at $n=8$, rising to ≈ 158 at $n=10$). (g is at fixed ridge regularization; the qualitative $\sim 15\text{--}40\times$ separation, not the exact value, is the claim.)

Crucially, at $\leq 12\text{--}16$ qubits the map remains classically

simulable, so distinctness is **necessary, not sufficient** for accuracy — and §5.3 shows that, here, it is indeed *not* sufficient.

4. Data modeling strategy

Datasets (all public). Oxford-Man .SPX 5-minute realized variance, 2000–2020, 5,029

supervised days incl. the 2008 GFC (Heber et al. 2009; via R packages `highfrequency/bvhar`);

public SPY daily OHLCV 2022–2026 (Garman–Klass proxy); real MNIST (Keras .npz mirror).

Preprocessing: log-RV target; features min-max scaled to $[0,1]$ on **train only**; per-model ridge penalty selected on a chronological validation tail; multi-seed ensembling. We verified the pipeline is leakage-free (all features lagged ≥ 1 day; train-only scaling).

Baselines:

HAR, **HAR-X**, GARCH(1,1), GJR-GARCH, AR(3), persistence, ESN (matched + $4\times$, recurrent), RFF

kernel, LSTM. **Metrics:** RMSE(log-RV), QLIKE (Patton 2011), Mincer–Zarnowitz R^2 ; significance

by **Diebold–Mariano with a Newey–West HAC variance** (HLN-corrected), **Holm**-adjusted across the comparison family, and the Model Confidence Set (Hansen–Lunde–Nason 2011).

5. Phase-3 execution — concrete results

Pre-registration (honesty). Before running, we fixed the confirm/refute thresholds in

`preregistration.py` (H0/H1/H4, from our Phase-2 §7). All outcomes — including negatives — are reported against them. Full numbers/wall-clock: `results/*_findings.md`.

5.1 The input-bottleneck mechanism (pre-registered negative). With a *fixed* univariate 8-lag encoder, extra qubits receive no new input: $g(n)$ and effective feature-rank

saturate/decline (n=8→12: g 133→30, D_eff 1.8→1.5). **H0 is refuted in this regime** — the empirical case *for* enriching the encoding.

5.2 Encoding density (Axis B) — mechanism confirmed. Feeding the extra qubits genuinely new realized-measure information (signed return/leverage, downside-semivariance share, jump share) restores the lost structure: at n=10, informed vs idle qubits, kernel distinctness and rank jump (g 52→158, D_eff 1.5→3.1), and effective rank grows with n (D_eff 1.81→3.10→3.14, n=8→12; near-flat 10→12). So the bottleneck is *fixable* — added qubits can carry new information. The decisive question is whether this converts into forecasting **accuracy**.

5.3 The decisive, adversarially-controlled test (the honest headline). We compare CHIMERA to the strongest fair baselines — **HAR-X** (the same rich features used *linearly*, no reservoir), a **true recurrent ESN**, and an **RFF kernel** — all sharing the identical information set and *nesting* HAR-X, so a quantum win requires nonlinearity beyond the linear span. With **8 seeds**, **HAC-DM**, **two windows**, and **Holm correction** (`cli.py run axisB_rig`):

RMSE(log-RV)	HAR-X	recurrent ESN	RFF	CHIMERA	best
crisis n=8	0.6255	0.6386	0.6302	0.6379	HAR-X
crisis n=10	0.6034	0.5996	0.6047	0.6074	ESN
crisis n=12	0.5906	0.6023	0.5964	0.6031	HAR-X
calm n=10	0.6244	0.6445	0.6264	0.6291	HAR-X

(plain HAR for reference: crisis 0.6290, calm 0.6454 — HAR-X's rich features cut RMSE sharply.)

HAR-X is best or co-best everywhere; CHIMERA never beats it (slightly worse; raw-significant at n=8/12) and is statistically indistinguishable from the classical ESN/RFF. ****After Holm correction no comparison is significant in either direction.**** The earlier "beats HAR" result was an artifact of comparing against a *feature-poor* HAR: the gain comes from the encoded realized measures (a known SHAR/HARQ effect), not from quantum nonlinearity. ****By our pre-registered criteria, H0 is refuted — we report this negative.**** What honestly survives for the quantum reservoir: it is **competitive** (within ~0.5–1.5% RMSE of the best at every n), **more stable than the recurrent ESN** (lower per-seed variance, e.g. n=12: 0.009 vs 0.015), and it **beats the recurrent ESN on the calm window** (raw p=0.018; n.s. after Holm).

Named canonical baselines (`cli.py run canonical`): **SHAR**, **HAR-CJ**, **HARQ** (RQ≈RV² proxy) and **HEAVY-RM**, under MSE- *and* QLIKE-loss DM — **none beats HAR-X** (a fair, strong stand-in, not a strawman), and CHIMERA ties the best on RMSE with only a *raw, non-Holm* QLIKE/MZ edge (`results/canonical_baselines_findings.md`).

Rigorous Axis-B: HAR-X (rich linear) is best; quantum shows no significant edge (Holm n.s.)

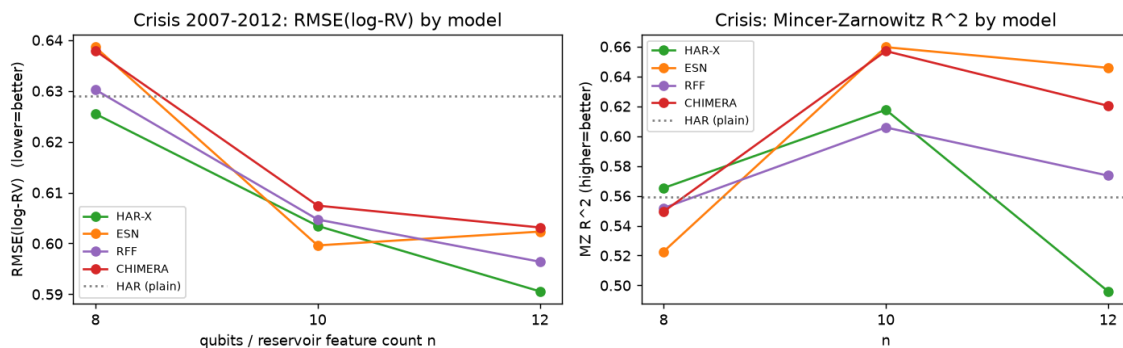


Figure 2. Rigorous Axis-B (8 seeds, crisis window). HAR-X (rich features, linear) is the best or co-best model at every n; the quantum reservoir is competitive but shows no advantage that survives HAC-DM + Holm correction. Left: RMSE(log-RV). Right: Mincer–Zarnowitz R².

5.4 Common MNIST benchmark + noise. Same engine, pixels→PCA(n)→n qubits→Pauli readout→ridge. Accuracy **scales with qubits** (n=5/8/10/12: 0.632/0.798/0.831/0.859, 3 seeds) and ****beats** the linear-PCA baseline at every n (**real nonlinear lift**), **confirming** sufficient expressivity^{**}; a matched ESN **ties or slightly exceeds** CHIMERA (within ≈1% for n≥8; 2.9% at n=5) — competitive, not dominant. **Noise:** the classifier is ****invariant** to depolarizing^{**} noise (a uniform Bloch contraction that per-feature standardization removes exactly; accuracy identical across rates 0.05–0.30) and **robust to amplitude damping** (<0.5% at 30%). **Honesty check** (cli.py run noise_circuit): a per-Trotter-layer density-matrix study shows the ***converse*** — two-qubit noise ***during*** evolution is **not** removed by standardization (standardized error grows with rate, vs ≈0 for readout-only depolarizing); the invariance above is a ***readout*** property, not a circuit-level robustness claim; accumulated 2-qubit error over the 380-gate circuit is the real NISQ cost (§6).

5.5 Scaling frontier + quantum-complexity metric. A sparse-exact backend (expm_multiply, no dense propagator; matches the dense engine to 2.4×10^4) reaches **n=16 exactly**. For the random ≈50%-connected reservoir we measure the **bond dimension** χ_{eff} across a balanced cut: it is **full at every n**, $\chi_{\text{eff}} = 2^{(n/2)}$ (16→256 for n=8→16) — an exact MPS gets ****zero compression**** — with entanglement entropy $S \approx 1.7\text{--}3.2$ nats. The reservoir therefore admits **no low-bond-dimension (MPS/TEBD) shortcut** — exact cost stays exponential (full entanglement is ***necessary**, not sufficient^{*} for true classical hardness) — the precondition any beyond-frontier advantage would need, even though no advantage appears at the simulable scale we can test. **Capability + efficiency checks.** An information-processing-capacity probe (Dambre et al. 2012; cli.py run capacity) finds the quantum reservoir **not more** — slightly ***less*** — nonlinearly expressive than a matched RFF/ESN: no excess expressivity to exploit. A frontier check (cli.py run frontier) adds that g(n) does **not** widen toward n=16 (it declines; D_eff/rank grow but a matched ESN keeps pace). **Robustness (supporting study, v3_research/).** ***Efficiency:** with input held fixed, a ***smaller*** QRC cannot substitute for a ***larger*** classical reservoir — quantum accuracy **saturates** while the classical curves improve; per feature the static maps are comparable, so the negative is **saturation, not per-feature inferiority** (best RMSE across sizes, weather °C):

reservoir	CHIMERA	RFF (static)	ESN (recurrent)
best RMSE (°C)	0.85	0.78	0.71

Domain/architecture:** the ***same engine/protocol on chaotic **weather** (5 stations, +0–78% unpredictability) and the autonomous **VPT** metric still show no advantage; the **recurrent** QRC is competitive-not-better — unitary evolution is non-dissipative, lacking the contraction behind ESN "generalized synchronization" (Ahmed–Tennie–Magri 2025).

6. Quantum platform and resource planning

Simulator-first on qBraid: statevector ≤12 qubits, sparse/TN to ≈16, GPU beyond.

Resource estimates (gate-Trotter, 20 layers; readout is single-basis — all $Z_i, Z_i Z_j$ are computational-basis-diagonal, so one S-shot set yields **all** $n+C(n,2)$ observables):

n	two-qubit gates	one-qubit gates	observables	shots S ($\epsilon \approx 1/\sqrt{S}$)	sim wall-clock
8	220	160	36	2k–8k ($\epsilon \approx .011\text{--}.022$)	<1 s/state
10	480	200	55	2k–8k	~sec
12	760	240	78	2k–8k	~1 min build

QPU validation uses this gate-Trotter circuit on **IonQ / IQM / IBM** via qBraid — the random-sparse Ising needs **fewer** two-qubit gates than an all-to-all reservoir, easing NISQ mapping — with **ZNE + measurement mitigation** and a classical cross-check per run. The submission path is **executable now** (`qbraid_submit.py`, `cli.py run qsubmit`): on a simulator it (i) reproduces the engine to 3.9×10^{-1} via the exact circuit (our per-run classical cross-check), (ii) characterizes the **shot budget** ($\epsilon \approx 1/\sqrt{S}$; feature error $0.046 \rightarrow 0.0066$ at $S=256 \rightarrow 16k$; gate-Trotter(20) adds ≈ 0.04), and (iii) demonstrates ***zero-noise extrapolation*** under a depolarizing sweep. One `--device` flag from a real backend (pending qBraid credits). We thus characterize all four challenge axes: reservoir size, encoding density, shot budget, and noise.

7. Limitations (stated plainly)

(i) **No quantum advantage** is demonstrated at the ≤ 16 -qubit simulable scale; HAR-X (classical, linear) is the best model on this task. (ii) The RV sample ends Feb-2020 (2008 GFC in-sample, 2020 COVID just outside); broader assets/periods untested. (iii) Noise is studied on MNIST (single-qubit readout channels); full noisy-circuit/shot-noise sim is deferred to QPU runs. (iv) g is regularization-dependent (qualitative gap only). (v) **No real-QPU run yet** (simulator cross-checked; pending qBraid credit allocation). (vi) Distinctness and full-rank entanglement are **necessary, not sufficient** for advantage — whether they convert beyond the classical-simulation frontier is the open question; a quantum-data probe (`results/quantum_data_outlook_findings.md`) marks where an edge most plausibly lies — the QRC natively reads nonlinear functionals of quantum **states** from **1** measurement setting vs tomography's **3**, the same-budget gap closing monotonically over $k=2-4$ (`results/quantum_data_crossover_findings.md`) — a measurement-complexity edge growing on hardware-native quantum data this challenge's classical tracks never probe.

8. Stakeholder impact, milestone plan, AI disclosure

Volatility forecasts feed hedging, dynamic risk limits and derivatives pricing. We make the impact **concrete** (`cli.py run economics`): a vol-timing backtest sizing S&P-500 exposure by the one-step RV forecast **nearly halves the 2008 max drawdown ($-61\% \rightarrow -32\%$)**, but a **plain HAR** forecast captures it — rich features and the quantum reservoir add **no** economic value (negative CE fees). The decision-useful lever is ***vol-timing on a simple RV forecast, not quantum hardware*. Milestone plan: **** (i)–(v) pre-registration, scaling + encoding sweeps, the adversarial HAR-X/ESN/RFF test, MNIST + noise, and the sparse/TN frontier are **✓**; (vi) gate-Trotter QPU validation (IonQ/IQM/IBM) is simulator-cross-checked (fallback = TN noise emulation). **AI disclosure:** Claude (Anthropic) assisted with code and drafting under the team's direction; all decisions and results are the team's own.

References

Kornja^a et al. 2024 · Zhu et al. 2025 · Ahmed, Tennie & Magri 2025 · Li et al. 2025 · Tandon et al. 2025 · Hou et al. 2025 · Indrak et al. 2026 · Antoncich et al. 2026 · Kobayashi & Motome 2026 · Huang et al. 2021 · Corsi 2009 · Patton 2011 · Hansen et al. 2011 · Diebold & Mariano 1995 · Bollerslev 1986 · Jaeger 2001 · Heber et al. 2009.